

WEEK 2 – BASIC ENCRYPTION & DECRYPTION

Introduction

Encryption is the process of converting readable data into encoded format to protect information from unauthorized access. Decryption converts encrypted data back into original readable form.

Encryption is widely used in:

- Banking systems
- Secure messaging
- Password protection
- Online transactions
- Secure communication

Objective

The objective of this project was to implement a basic encryption and decryption technique using Caesar Cipher.

Caesar Cipher Concept

Caesar Cipher shifts each alphabet character by a fixed number of positions.

Example:

A → D

B → E

C → F

Encryption Algorithm

1. Accept user input
2. Choose shift value
3. Shift characters forward
4. Generate encrypted text

Decryption Algorithm

1. Accept encrypted text
 2. Reverse shift operation
 3. Restore original message
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Applications of Encryption

- Secure communication
- Data protection
- Password security
- Confidential messaging
- Digital privacy

Code

Input

```
text = input("Enter Text: ")  
shift = 3
```

```
encrypted = ""
```

```
for char in text:  
    encrypted += chr(ord(char) + shift)
```

```
print("Encrypted Text:", encrypted)
```

```
decrypted = ""
```

```
for char in encrypted:  
    decrypted += chr(ord(char) - shift)
```

```
print("Decrypted Text:", decrypted)
```

```
print("Encryption Process Completed")
```

output

```
Enter Text: Encrypted Text: Koor#Zruog  
Decrypted Text: Hello World  
Encryption Process Completed
```